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MTH645

Fuzzy Logic and Applications

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Lecture # 25

Example: $x \Delta y = xy$, $f(x) = e^{-\left(\frac{1-x}{x}\right)}$, $x \Delta_f y = ?$

Sol: $x \Delta_f y = f^{-1}\left((f(x) \Delta f(y)) \vee a\right)$

let $y = e^{-\left(\frac{1-x}{x}\right)}$

$\ln y = \ln e^{-\left(\frac{1-x}{x}\right)} = -\left(\frac{1-x}{x}\right) \cdot \ln e \quad \because \ln e = 1$

$\ln y = -\left(\frac{1-x}{x}\right)$

$x \cdot \ln y = x - 1$

$1 = x - x \ln y = x(1 - \ln y)$

$\Rightarrow x = \frac{1}{1-\ln y} = f^{-1}(y)$

$\Rightarrow f^{-1}(x) = \frac{1}{1-\ln x}$

$f(0) = e^{-\left(\frac{1-0}{0}\right)}$

$= e^{-\infty} = \frac{1}{e^{\infty}}$

$f(0) = \frac{1}{\infty} = 0$

$\Rightarrow a = 0$

$x \Delta_f y = f^{-1}\left((f(x) \Delta f(y)) \vee a\right)$

$= f^{-1}\left((f(x) \cdot f(y)) \vee a\right)$

$= f^{-1}\left((f(x) \cdot f(y)) \vee 0\right)$

$= f^{-1}\left(f(x) \cdot f(y)\right)$

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$$x \Delta_f y = f^{-1}\left[e^{-\left(\frac{1-x}{x}\right)} \cdot e^{-\left(\frac{1-y}{y}\right)}\right]$$

$$= f^{-1}\left[e^{\left[-\left(\frac{1-x}{x}\right)-\left(\frac{1-y}{y}\right)\right]}\right]$$

$$= \frac{1}{1-\ln\left[e^{\left[-\left(\frac{1-x}{x}\right)-\left(\frac{1-y}{y}\right)\right]}\right]}$$

$$= \frac{1}{1-\left[-\left(\frac{1-x}{x}\right)-\left(\frac{1-y}{y}\right)\right]}$$

$$= \frac{1}{1+\frac{1-x}{x}+\frac{1-y}{y}} = \frac{1}{\frac{xy+y(1-x)+x(1-y)}{xy}}$$

$$= \frac{xy}{xy+y-xy+x-xy}$$

$$x \Delta_f y = \frac{xy}{x+y-xy} = \Delta_4$$

Lecture #24

Example: $f(x) = e^{x-1}$, $x \Delta y = xy$, $x \Delta_f y = ?$

Sol: let $y = e^{x-1}$

$\ln y = x - 1$

$x = 1 + \ln y = f^{-1}(y)$

$\Rightarrow f^{-1}(x) = 1 + \ln x$

$f(0) = e^{0-1} = e^{-1}$

$a = f(0) = \frac{1}{e}$

Now $x \Delta_f y = f^{-1}((f(x) \Delta f(y)) \vee a)$

$= f^{-1}\left((f(x) f(y)) \vee \frac{1}{e}\right)$

$= f^{-1}\left((e^{x-1} \cdot e^{y-1}) \vee \frac{1}{e}\right)$

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$$x \Delta_f y = f^{-1}\left(e^{x+y-2} \vee \frac{1}{e}\right)$$

$$= f^{-1}\left(e^{x+y-2} \vee e^{-1}\right)$$

$$= 1 + \ln\left(e^{x+y-2} \vee e^{-1}\right)$$

$$= 1 + \left(\ln e^{x+y-2} \vee \ln e^{-1}\right)$$

$$= 1 + (x+y-2 \vee -1)$$

$$= 1 + x + y - 2 \vee 1 - 1$$

$$x \Delta_f y = x + y - 1 \vee 0 = \Delta_1$$

Example: $f(x) = \frac{x}{2-x}$, $x \Delta y = xy$ $\quad x \Delta_f y = ?$

Sol: let $y = \frac{x}{2-x}$

$$y(2-x) = x$$

$$2y - xy = x$$

$$2y = x + xy$$

$$2y = x(1+y)$$

$$\Rightarrow x = \frac{2y}{1+y} = f^{-1}(y)$$

$$\Rightarrow f^{-1}(x) = \frac{2x}{1+x}$$

$$f(0) = \frac{0}{2-0} = 0$$

$$\Rightarrow a = 0$$

$$x \Delta_f y = f^{-1}\left((f(x) \Delta f(y)) \vee a\right)$$

$$= f^{-1}\left(f(x) \cdot f(y) \vee 0\right)$$

$$= f^{-1}\left(f(x) \cdot f(y)\right)$$

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$$= f^{-1}\left(\frac{x}{2-x} \cdot \frac{y}{2-y}\right)$$

$$= f^{-1}\left(\frac{xy}{(2-x)(2-y)}\right)$$

$$= f^{-1}\left(\frac{xy}{4-2x-2y+xy}\right)$$

$$= \frac{2\left(\frac{xy}{4-2x-2y+xy}\right)}{1+\frac{xy}{4-2x-2y+xy}}$$

$$= \frac{\frac{2xy}{4-2x-2y+xy}}{\frac{1(4-2x-2y+xy)+xy}{4-2x-2y+xy}} = \frac{2xy}{4-2x-2y+xy+xy}$$

$$= \frac{2xy}{4-2x-2y+2xy} = \frac{2xy}{2(2-x-y+xy)}$$

$$x \Delta_f y = \frac{xy}{2-x-y+xy} = \frac{xy}{2-(x+y-xy)} = \Delta_2$$

Lecture # 25

Nilpotent t-norm:

Δ is nilpotent if for $b \neq 1$, $b^{[n]} = 0$, for some +ve integer "n"; where "n" depends on b.

Strict t-norm:

Δ is strict if for $b \neq 0$, $b^{[n]} > 0$, for some +ve integer "n";

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For Example $b^{[3]} = b \Delta b \Delta b$

Exercise:

Q: prove that Δ_1 is nilpotent t-norm.

Sol: $x \Delta_1 y = \max(x+y-1, 0)$

let $x \neq 1$

$\Rightarrow x \Delta_1 x = \max(x+x-1, 0) = \max(2x-1, 0)$

if $x \leq \frac{1}{2} \Rightarrow x \Delta_1 x = 0$

if $1 > x > \frac{1}{2} \Rightarrow x > 2x-1$

$x^{[3]} = x \Delta_1 x \Delta_1 x = \max(\max(2x-1, 0) + x - 1, 0)$

$= \max(2x-1+x-1, 0) = \max(3x-2, 0)$

$\Rightarrow 3x-2 < 2x-1 < x$

$\Rightarrow 4x-3 < 3x-2 < 2x-1 < x$

$\cdots\cdots < nx-(n-1) < (n-1)x-(n-2) < \cdots\cdots < x$

$\Rightarrow x^{[n]} = \max(nx-(n-1), 0)$

So we get +ve n such that $nx-(n-1)$ becomes -ve.

$\Rightarrow x^{[n]} = 0$

$\Rightarrow \Delta_1$ is nilpotent t-norm.

Q: prove that Δ_0 is nilpotent t-norm.

Sol: $x \Delta_0 y = \begin{cases} \min(x,y) & ; \text{if } \max(x,y) = 1 \\ 0 & ; \text{if otherwise} \end{cases}$

Let $\max(x,y) = 1$

$\Rightarrow x \neq 1 \Rightarrow \max(x,x) \neq 1$

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$\exists$ there exists

$\Rightarrow x \Delta_0 x = 0$

$\Rightarrow x^{[2]} = 0$,

$\Rightarrow \Delta_0$ is nilpotent t-norm.

Zero Divisor:

A fuzzy number $a \in (0,1)$ is called zero divisor of t-norm Δ, if there exists $b \in (0,1)$ such that $a \Delta b = 0$

Theorem: If Δ is strictly monotone, then it has no zero divisor.

proof: let Δ is strictly monotone, we have to show that Δ has no zero divisor.

If $x > 0$ and $y < z$

$\Rightarrow x \Delta y < x \Delta z$

let $a \in (0,1)$ is zero divisor of Δ

If there exist $b \in (0,1)$ such $a \Delta b = 0$

$\Rightarrow a \Delta b = 0$, $a, b \in (0,1)$

Take $\frac{b}{2} < b$, $a \leq a$

$$a \Delta \frac{b}{2} \leq a \Delta b = 0$$

$$a \Delta \frac{b}{2} \leq 0$$

$a \Delta \frac{b}{2} = 0$ (because two fuzzy numbers cannot -ve)

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which is contradiction to strictly monotone.

=) Δ has no zero divisor.

Lecture # 26

Strictly monotone t-norm:

A t-norm Δ is called strictly monotone t-norm if $x \Delta y < x \Delta z$ whenever $x > 0$ and $y < z$

Cancelation Law in t-norm:

① If $x \Delta y = x \Delta z$ =) $x = 0$ or $y = z$

② If $x \Delta y = x \Delta z$ and $x = 0$ =) $y = z$

Theorem: Δ is strictly monotone t-norm if and only if it satisfies the cancelation law.

proof: let Δ is strictly monotone t-norm and we show that cancelation law holds.

Take $x \Delta y = x \Delta z \longrightarrow ①$

Suppose on contrary $x \neq 0$ and $y \neq z$

=) $x > 0$, let $y < z$

=) $x \Delta y < x \Delta z \longrightarrow ②$

From ① and ② =) $x \Delta y < x \Delta z$

=) which is contradiction to cancelation law

So, If Δ is strictly monotone t-norm then cancelation law holds.

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Conversly, If it satisfies cancelation law
then it is strictly monotone:
we have to show that
If $x \Delta y < x \Delta z$ whenever $x > 0$ and $y < z$.
let Δ is not strictly monotone t-norm.
$x \Delta y \geq x \Delta z$, whenever $x > 0$ and $y < z$.
$\rightarrow$ ①
Now take $x \leq x$ and $y \leq z$
$\Rightarrow x \Delta y \leq x \Delta z \longrightarrow$ ②
From ① and ② $\Rightarrow x \Delta y = x \Delta z$
which is contradiction to cancelation law.
$\Rightarrow x \Delta y < x \Delta z$, $x > 0$ and $y < z$
So Δ is strictly monotone.
which complete the proved.

Idempotent Element:

Let $a \in (0,1)$ is idempotent element
if $a \Delta a = a$
For example: (i) $0 \Delta 0 = 0 \Rightarrow 0$ is idempotent element
(ii) $1 \Delta 1 = 1 \Rightarrow 1$ is idempotent element.

Theorem: If T is strictly monotone then
it has only trivial idempotent element.
proof: let T has non trivial idempotent element
x and y.

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Let $x, y \in (0,1)$

$x \Delta y \leq x \Delta z \; ; \; x > 0 \; , y \leq z$

Take $x \leq x$ and $x < 1$

$\Rightarrow x \Delta x \leq x \Delta 1 = x$

$\Rightarrow x \Delta x \leq x$

which is contradiction.

So T is trivial idempotent element.

Lecture #27

Binary Fuzzy Relation:

A binary fuzzy relation is $R: U \times U \rightarrow [0,1]$.

(i) Reflexive Relation:

A binary fuzzy relation "R" is reflexive if $R(x,x) = 1$

(ii) Symmetric Relation:

A fuzzy binary relation "R" is symmetric if

$R(x,y) = 1 \Rightarrow R(y,x) = 1$

(iii) Transitive Relation:

~~Transit~~ A fuzzy binary relation R is transitive

if $R(x,y) = 1, R(y,z) = 1 \Rightarrow R(x,z) = 1$

where $x, y, z \in$ fuzzy numbers.

(iv) R is equivalance relation if R is reflexive, symmetric and transitive.

(v) Antisymmetric Relation:

A fuzzy binary relation R is antisymmetric

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if $R(x,y) = R(y,x) = 1 \Rightarrow x = y$.

More Generalized Definitions:

(ii) Reflexive Relation:

A fuzzy binary relation R is reflexive if $R(u,u) = 1$

(iii) Symmetric Relation:

A fuzzy binary relation R is symmetric if $R(u,v) = R(v,u)$.

(iii) Transitive Relation:

A fuzzy binary relation R is transitive if $R(u,w) \geq R(u,v) \wedge R(v,w)$.

(iv) Equivalance Relation:

A fuzzy binary relation R is equivalance if R is reflexive, symmetric and transitive relation.

(v) Antisymmetric Relation:

A fuzzy binary relation R is antisymmetric if $R(u,v) > 0$ and $R(v,u) > 0 \Rightarrow u = v$.

α-cut in fuzzy binary relation:

α-cut in fuzzy binary relation is

$$R_\alpha = R^{-1}([\alpha, 1]) = [(u,v) : R(u,v) \geq \alpha]$$

for example: $\alpha = 0.5$, $R = [((x_1,y_1), 0.3), ((x_1,y_2), 0.7), ((x_2,y_1), 0.9)]$

$$R_\alpha = R_{0.5} = R[((x_1,y_2), 0.7), ((x_2,y_1), 0.9)]$$

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Theorem: If R is a fuzzy on a set U, then R is a equivalance relation if and only if each α-cut, R_α is an equivalance relation on U.

Proof: Let R_α be a fuzzy equivalance relation.

Then (i) $R(u,u) = 1 \geq \alpha \Rightarrow (u,u) \in R_\alpha$, R_α is reflexive.

(ii) If $R(u,u) = 1 = R(v,u) \Rightarrow (u,v) \in R_\alpha$

$\Rightarrow R(u,v) = 1 \geq \alpha$ also $R(v,u) = 1 \geq \alpha$

$\Rightarrow (v,u) \in R_\alpha \Rightarrow R_\alpha$ is symmetric.

(iii) If $(u,v), (v,w) \in R_\alpha$

$\Rightarrow R(u,v) = 1 \geq \alpha$ and $R(v,w) = 1 \geq \alpha$

$\Rightarrow R(u,w) \geq R(u,v) \wedge R(v,w) = 1 \geq \alpha$

$R(u,w) \geq \alpha \Rightarrow R(u,w) \in R_\alpha \Rightarrow R_\alpha$ is transitive.

from (i), (ii) and (iii)

$\Rightarrow R_\alpha$ is an equivalance relation.

Conversly: let R_α is equivalme relation.

Then (i) If $(u,u) \in R_\alpha \Rightarrow R(u,u) \geq \alpha$ (put $\alpha = 1$)

$\Rightarrow R(u,u) \geq 1$

A fuzzy relation is not greater then 1.

So $R(u,u) = 1 \Rightarrow R$ is reflexive.

(ii) $\because (u,v), (v,u) \in R_\alpha$

$\Rightarrow R(u,v) \geq \alpha$ and $R(v,u) \geq \alpha$, $\forall \alpha \in (0,1]$

Take $\alpha = 1$, $R(u,v) \geq 1$ and $R(v,u) \geq 1$

A fuzzy numbers is not greater then 1.

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So $R(u,v) = 1$ and $R(v,u) = 1$

$\Rightarrow$ R is symmetric.

(iii) If $(u,v), (v,w) \in R_\alpha \Rightarrow (u,w) \in R_\alpha$

$\Rightarrow R(u,v) \geq \alpha$ and $R(v,w) \geq \alpha \Rightarrow R(u,w) \geq \alpha$

$\Rightarrow R(u,v) \wedge R(v,w) \geq \alpha$,

Take $\alpha = 1$, $\Rightarrow R(u,v) \wedge R(v,w) \geq 1$

The fuzzy numbers are not greater than 1.

So $R(u,v) \wedge R(v,w) = 1$ and $R(u,w) = 1$

From (i), (ii) and (iii). $\Rightarrow$ R is transitive.

$\Rightarrow$ R is equivalance relation.

Lecture # 28

Composition of Relation:

Let R and S be fuzzy relation in $u \times v$ and $v \times w$ respectively and let Δ be a t-norm. The composition $R \circ S$ of R and S with respect to Δ is the fuzzy relation on $u \times w$ with membership function

$$(R \circ S)(u,w) = \bigvee_{v \in V} [R(u,v) \Delta S(v,w)]$$

$x \Delta y = x \wedge y \Rightarrow$ max-min composition.

$x \Delta y = xy \Rightarrow$ max-product composition.

Example. Find the max-min and max product composition of R and S.

Sol:

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$$R = \begin{matrix} & y_1 & y_2 & y_3 \\ x_1 \\ x_2 \\ x_3 \end{matrix}\begin{bmatrix} 0.1 & 0.0 & 0.6 \\ 0.2 & 0.3 & 1 \\ 0 & 0.4 & 0.5 \end{bmatrix} \quad \text{and} \quad S = \begin{matrix} & s_1 & s_2 & s_3 \\ r_1 \\ r_2 \\ r_3 \end{matrix}\begin{bmatrix} 0.2 & 0.5 & 0 \\ 0.9 & 1 & 0.3 \\ 1 & 0.5 & 0.4 \end{bmatrix}$$

max-min composition:

$$R \circ S = \begin{bmatrix} \underline{0.6} & \underline{0.5} & 0.4 \\ 1 & 0.5 & 0.4 \\ 0.5 & 0.5 & 0.4 \end{bmatrix}$$

$\min(0.1, 0.2) = 0.1$
$\min(0.0, 0.9) = 0.0$ } $\max = \underline{0.6}$
$\min(0.6, 1) = 0.6$

$\min(0.1, 0.5) = 0.1$
$\min(0.0, 1) = 0.0$ } $\max = \underline{0.5}$
$\min(0.6, 0.5) = 0.5$

So on.

max-product composition:

$$R \circ S = \begin{bmatrix} 0.1 & 0 & 0.6 \\ 0.2 & 0.3 & 1 \\ 0 & 0.4 & 0.5 \end{bmatrix} \quad \begin{bmatrix} 0.2 & 0.5 & 0 \\ 0.9 & 1 & 0.3 \\ 1 & 0.5 & 0.4 \end{bmatrix}$$

$$R \circ S = \begin{bmatrix} 0.6 & 0.38 & 0.24 \\ 1 & 0.5 & 0.4 \\ 0.5 & 0.4 & 0.2 \end{bmatrix}$$

$0.1 \times 0.2 = 0.02$
$0 \times 0.9 = 0$ } $\max = 0.6$
$0.6 \times 1 = 0.6$

$0.1 \times 0.5 = 0.05$
$0 \times 1 = 0$ } $\max = 0.3$
$0.6 \times 0.5 = 0.38$

So on.

Example:

$R \circ R \subseteq R \iff R$ is transitive.

$\text{Max}(\text{Min}(R(u,v), R(v,w))) \leq R(u,w)$

Fuzzy Relation: Example:

$A = \{x_1, x_2\}$, $B = \{y_1, y_2, y_3\}$

$$A \times B = \left\{ \frac{(x_1, y_1)}{0.2}, \frac{(x_1, y_2)}{0.3}, \frac{(x_1, y_3)}{1}, \frac{(x_2, y_1)}{0}, \frac{x_2, y_2}{0.5}, \frac{x_2, y_3}{0.7} \right\}$$

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$$R_1 = \begin{array}{c} \\ x_1 \\ x_2 \end{array}\begin{array}{c} \begin{array}{ccc} y_1 & y_2 & y_3 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \end{array} \Rightarrow R_1 = \{(x_1, y_1), (x_2, y_2)\}$$

$\tilde{F}: U^2 \longrightarrow [0,1]$,

$$\tilde{R} = \begin{array}{c} \\ x_1 \\ x_2 \end{array}\begin{array}{c} \begin{array}{ccc} y_1 & y_2 & y_3 \end{array} \\ \begin{bmatrix} 0.2 & 0.3 & 1 \\ 0 & 0.5 & 0.7 \end{bmatrix} \end{array}$$

$\tilde{R} = A \times B \longrightarrow [0,1]$, $\tilde{R} = \{((x_1, y_1), 0.2), ((x_2, y_2), 0.5)\}$

Complete Relation of $A \times B$ if

$$\tilde{R} = \begin{array}{c} \\ x_1 \\ x_2 \end{array}\begin{array}{c} \begin{array}{ccc} y_1 & y_2 & y_3 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \end{array}$$

Null Relation of $A \times B$ if

$$\tilde{R} = \begin{array}{c} \\ x_1 \\ x_2 \end{array}\begin{array}{c} \begin{array}{ccc} y_1 & y_2 & y_3 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

Example:

$$R = \begin{bmatrix} 0.2 & 0.3 & 0.7 & 0.9 \\ 0.5 & 0.2 & 0.3 & 0.7 \\ 0.1 & 0.3 & 0.9 & 0.8 \end{bmatrix}, \quad S = \begin{bmatrix} 0 & 0.7 & 0.1 & 0.8 \\ 0.9 & 0.2 & 0.7 & 0.1 \\ 0 & 0.3 & 0.9 & 1 \end{bmatrix}$$

Union:

$$R \cup S = \begin{bmatrix} 0.2 & 0.7 & 0.7 & 0.9 \\ 0.9 & 0.2 & 0.7 & 0.7 \\ 0.1 & 0.3 & 0.9 & 1 \end{bmatrix}$$

$$R \cup S = \max(R, S)$$

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Intersection:

$$R \cap S = \begin{bmatrix} 0 & 0.3 & 0.1 & 0.8 \\ 0.5 & 0.2 & 0.3 & 0.1 \\ 0 & 0.3 & 0.9 & 0.8 \end{bmatrix}$$

$$\boxed{R \cap S = \min(R, S)}$$

Compliment.

$$R^c = 1 - R = \begin{bmatrix} 0.8 & 0.7 & 0.3 & 0.1 \\ 0.5 & 0.8 & 0.7 & 0.3 \\ 0.9 & 0.7 & 0.1 & 0.2 \end{bmatrix}$$

$$\boxed{R^c = 1 - R}$$

Example:

$$\tilde{S} = \begin{bmatrix} 0 & 0.7 & 0.1 & 0.8 \\ 0.9 & 0.2 & 0.7 & 0.1 \\ 0 & 0.3 & 0.9 & 1 \end{bmatrix}$$

Find α-cut of $\tilde{S}$

let $\alpha = 0.4$

$$\tilde{S}_\alpha = \tilde{S}_{0.4} = \begin{bmatrix} 0 & 0.7 & 0 & 0.8 \\ 0.9 & 0 & 0.7 & 0 \\ 0 & 0 & 0.9 & 1 \end{bmatrix}$$

$$\boxed{\tilde{S}_\alpha = \tilde{S}(u,v) \geq \alpha}$$

Lecture #29

Fuzzy Implications:

A Fuzzy implications In a map $\Rightarrow : [0,1]^2 \rightarrow [0,1]$ satisfying

$\Rightarrow$	1	0
1	1	0
0	1	1

$(1 \Rightarrow 1) = 1$

$(0 \Rightarrow 1) = 1$

$(0 \Rightarrow 0) = 1$

$(1 \Rightarrow 0) = 0$

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$(x \Rightarrow y) = x$ implication y

R-Implication:

A R-implication is a map $\Rightarrow : [0,1]^2 \to [0,1]$ such that $(x \Rightarrow y) = \max\{z \in [0,1] : x \Delta z \leq y\}$

$= \vee\{z \in [0,1] : x \Delta z \leq y\}$

(i) $(1 \Rightarrow 1) = \max\{z \in [0,1] : 1 \Delta z \leq 1\}$

$= \max\{z \in [0,1] : z \leq 1\}$

$= \max\{z \in [0,1]\} = 1$

$(1 \Rightarrow 1) = 1$

(ii) $(1 \Rightarrow 0) = \max\{z \in [0,1] : 1 \Delta z \leq 0\}$

$= \max\{z \in [0,1] : z \leq 0\}$

$= \max\{0\}$

$(1 \Rightarrow 0) = 0$

(iii) $(0 \Rightarrow 0) = \max\{z \in [0,1] : 0 \Delta z \leq 0\}$

$= \max\{z \in [0,1]\}$

$(0 \Rightarrow 0) = 1$

(iv) $(0 \Rightarrow 1) = \max\{z \in [0,1] : 0 \Delta z \leq 1\}$

$= \max\{z \in [0,1]\}$

$(0 \Rightarrow 1) = 1$

$(x \Rightarrow y) = \max\{z \in [0,1] : z \Delta x \leq y\}$

If $x \leq y$; $z \leq 1$

$\Rightarrow x \Delta z \leq y \Delta 1 = y$

$\Rightarrow x \Delta z \leq y$

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Example 1: $x \Delta y = \min(x,y) = x \wedge y$

$$(x \Rightarrow y) = \max\{ z \in [0,1] : \min(x, \underbrace{x \Delta z}_{\leq y}) \leq y \}$$

$$= \begin{cases} 1 & ; x \leq y \\ y & ; x > y \end{cases}$$

Example 2: $x \Delta y = xy$

$$(x \Rightarrow y) = \max\{ z \in [0,1] : \min(x, \underbrace{x \Delta z}_{\leq y}) \leq y \}$$

$$= \begin{cases} 1 & ; x \leq y \\ y/x & ; x > y \end{cases}$$

Example 3: $x \Delta y = \max(0, x+y-1)$

$$(x \Rightarrow y) = \max\{ z \in [0,1] : x \Delta z \leq y \}$$

$$= \max\{ z \in [0,1] : \max(0, x+z-1) \leq y \}$$

$$(x \Rightarrow y) = \begin{cases} 1 & ; x \leq y \\ y - x + 1 & ; x > y \end{cases}$$

Theorem: Let Δ be archemidian t-norm and f is generator of t-norm. Then $(x \Rightarrow y) = f^{-1}\left[\frac{f(y)}{f(x)} \wedge 1\right]$

proof: $x \Delta y = f^{-1}(f(x) \cdot f(y) \vee f(0))$

$$(x \Rightarrow y) = \max\{ z \in [0,1] : x \Delta z \leq y \}$$

$$= \max\{ z \in [0,1] : f^{-1}(f(x) f(z) \vee f(0)) \leq y \}$$

$f : [0,1] \to [a,1] \Rightarrow 0 \leq a < 1$

Now $f^{-1}(f(x) f(z) \vee f(0)) \leq y$

$\Rightarrow f(x) f(z) \vee f(0) \leq f(y)$

$\Rightarrow f(x) f(z) \vee a \leq f(y)$

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∇ Nebla η Eta

$\Rightarrow$ $f(x) f(z) \leq f(y)$

$\Rightarrow$ $f(z) \leq f(y)/f(x)$

$\Rightarrow$ $z \leq f^{-1}\left(\frac{f(y)}{f(x)} \wedge 1\right)$

$(x \Rightarrow y) = F^{-1}\left(\frac{f(y)}{f(x)} \wedge 1\right)$

(proved)

Lecture #30

∇- Implication:

A ∇- implication is a map $\Rightarrow [0,1]^2 \rightarrow [0,1]$ of the form $(x \Rightarrow y) = \eta(x) \nabla y$; $\eta(x) = 1 - x$.

∇ is t-conorm.

Example 1: $x \nabla y = \max(x,y) = \vee(x,y) = x \vee y$

Sol: $(x \Rightarrow y) = \eta(x) \nabla y$

$= (1-x) \nabla y = \max(1-x, y)$

Example 2: $x \nabla y = x + y - xy$

Sol: $(x \Rightarrow y) = \eta(x) \nabla y$

$= (1-x) \nabla y$

$= (1-x) + y - (1-x)(y)$

$= 1 - x + y - y + xy$

$(x \Rightarrow y) = 1 - x + xy$

Example 3: $x \nabla y = 1 \wedge (x+y)$

Sol: $(x \Rightarrow y) = \eta(x) \nabla y = (1-x) \nabla y$

$= 1 \wedge (1-x+y) = \min(1, 1-x+y)$

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$(0 \Rightarrow 0) = 1$

$(1 \Rightarrow 1) = 1$

$(0 \Rightarrow 1) = 1$

$(1 \Rightarrow 0) = \eta(x) \nabla y = (1-1) \nabla 0 = 0 \nabla 0$

$(1 \Rightarrow 0) = 0$

Example: $g(x) = e^{-x^2}$ $\quad \eta(x) = \sqrt{1-x^2}$

Then $(x \Rightarrow y) = \begin{cases} 1 & ; \ x \le y \\ \sqrt{1-x^2+y^2} & ; \ x > y \end{cases}$

Sol: $(x \Rightarrow y) = \eta(x) \nabla y$

$= g^{-1}(g(\eta(x)) g(y) \vee g(1))$

$= g^{-1}(g(\sqrt{1-x^2}) e^{-y^2} \vee e^{-1})$

$= g^{-1}(e^{-(1-x^2)} e^{-y^2} \vee e^{-1})$

$= g^{-1}(e^{-1+x^2-y^2} \vee e^{-1})$

$(x \Rightarrow y) = \sqrt{\ln(e^{-1+x^2-y^2})^{-1}} \vee \sqrt{\ln(e^{-1})^{-1}}$

$= \sqrt{\ln e^{1-x^2+y^2}} \vee \sqrt{\ln e^1}$

$= \sqrt{(1-x^2+y^2)\ln e} \vee \sqrt{\ln e}$

$= \sqrt{1-x^2+y^2} \vee \sqrt{1}$

$(x \Rightarrow y) = \sqrt{1-x^2+y^2} \vee 1$

$y = e^{-x^2}$

$\ln y = \ln e^{-x^2}$

$\ln y = -x^2$

$x^2 = -\ln y$

$x^2 = \ln y^{-1}$

$g^{-1}(y) = x = \sqrt{\ln y^{-1}}$

$$(x \Rightarrow y) = \begin{cases} 1 & ; \ x \le y \\ \sqrt{1-x^2+y^2} & ; \ x > y \end{cases}$$

$\because \boxed{\ln e = 1}$

(proved)

Lecture #31

Distance Between Two points:

Let $A(x_1, y_1)$ and $B(x_2, y_2)$, then the distance D between two points is calculated as:

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

for example: $A(-2, 1)$ and $B(2, 4)$, then the D is

$$D = \sqrt{(2-(-2))^2 + (4-1)^2} = \sqrt{4^2 + 3^2}$$

$$= \sqrt{16+9} = \sqrt{25}$$

$$D = 5$$

Distance Between Fuzzy Sets:

Let $X = \{x_1, x_2, \dots, x_n\}$ be a discourse set the fuzzy sets (FSs) A on X is represented in terms of a function $m : X \longrightarrow [0, 1]$ such as

$$A = \{\langle x_i, m_A(x_i)\rangle \mid x_i \in X\}$$

where $m_A(x_i)$ is called a membership function and always bounded in the closed interval $[0, 1]$.

Let A and B two fuzzy sets on $X = \{x_1, x_2, \dots, x_n\}$, then the most widely used distances for fuzzy sets are:

(1) Euclidean distance $d_e(A, B)$

$$d_e(A, B) = \sqrt{\sum_{i=1}^{n} \left(m_A(x_i) - m_B(x_i) \right)^2}$$

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(2) Normalized Euclidean distance $d_e^n(A,B)$

$$d_e^n(A,B) = \frac{1}{n}\sqrt{\sum_{i=1}^{n}\left(m_A(x_i) - m_B(x_i)\right)^2}$$

(3) Hamming distance $d_h(A,B)$

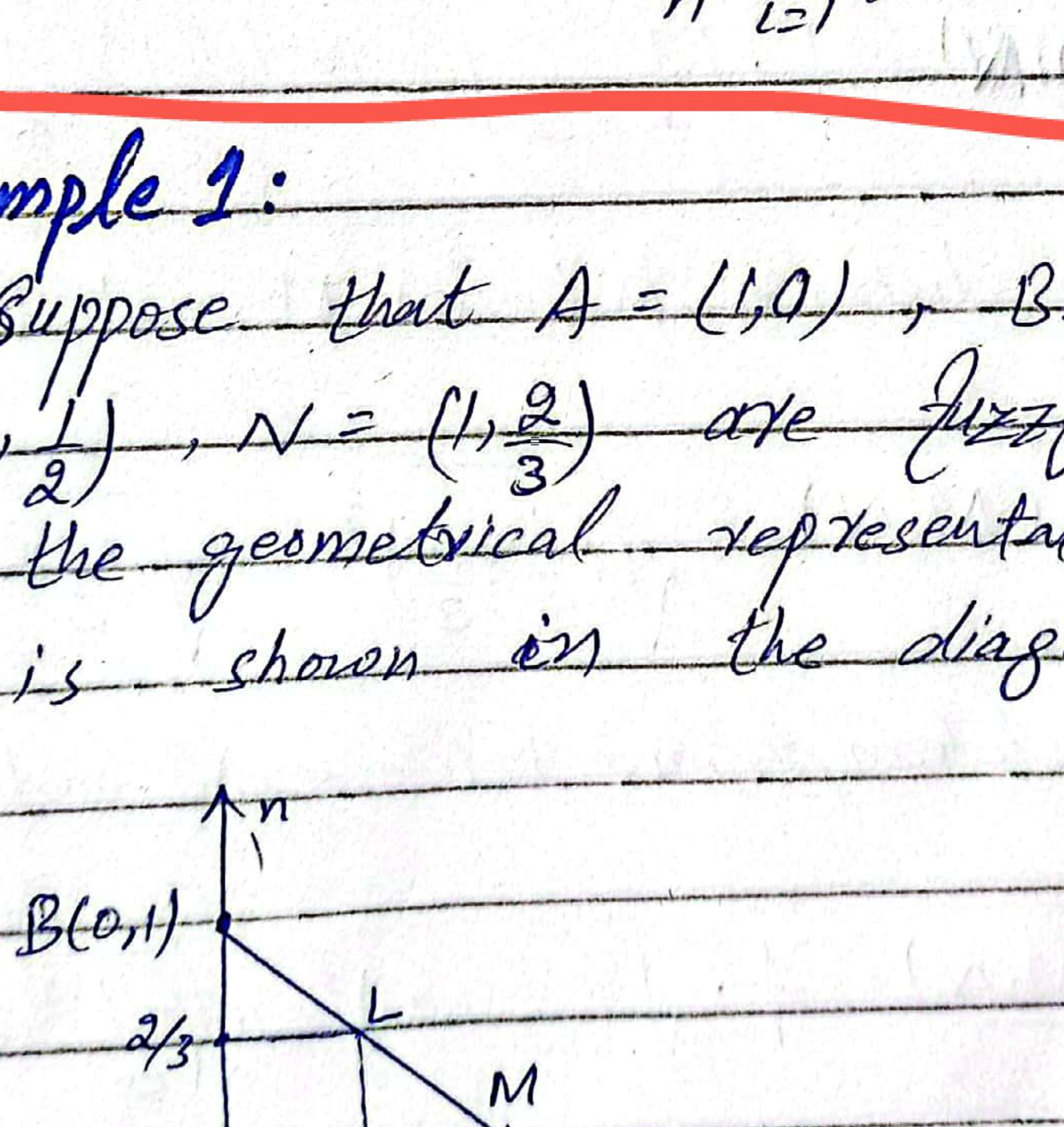
$$d_h(A,B) = \sum_{i=1}^{n} \left| m_A(x_i) - m_B(x_i) \right|$$

(4) Normalized Hamming distance: $d^n(A,B)$

$$d^n(A,B) = \frac{1}{n}\sum_{i=1}^{n} \left| m_A(x_i) - m_B(x_i) \right|$$

Example 1:

Suppose that $A = (1,0)$, $B = (0,1)$, $L = (1, \frac{1}{3})$, $M = (1, \frac{1}{2})$, $N = (1, \frac{2}{3})$ are fuzzy sets on $X = \{1\}$ then the geometrical representation of these sets is shown in the diagram below:



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(1) Therefore, the Hamming distance between the fuzzy sets L and M is:

$$d(L,M) = \left|\frac{2}{3} - \frac{1}{3}\right| = \frac{1}{3}$$

Similarly, the Hamming distance between the fuzzy sets M and N is:

$$d(M,N) = \left|\frac{1}{2} - \frac{2}{3}\right| = \left|\frac{-1}{6}\right| = \frac{1}{6}$$

(2) Now the normalized Hamming distance between the fuzzy sets M and N is

$$d^n(M,N) = \frac{1}{1} \cdot \left|\frac{1}{2} - \frac{2}{3}\right| = 1 \cdot \left|\frac{-1}{6}\right| = \frac{1}{6}$$

(3) Euclidean distance between M and N is:

$$d_e(M,N) = \sqrt{\left(\frac{1}{2} - \frac{2}{3}\right)^2} = \sqrt{\left(\frac{-1}{6}\right)^2} = \frac{1}{6}$$

(4) Normalized Euclidean distance between M and N is:

$$d_e^n(M,N) = \frac{1}{1} \cdot \sqrt{\left(\frac{1}{2} - \frac{2}{3}\right)^2} = 1 \cdot \sqrt{\left(\frac{-1}{6}\right)^2} = \frac{1}{6}$$

Lecture #32:

Example 2:

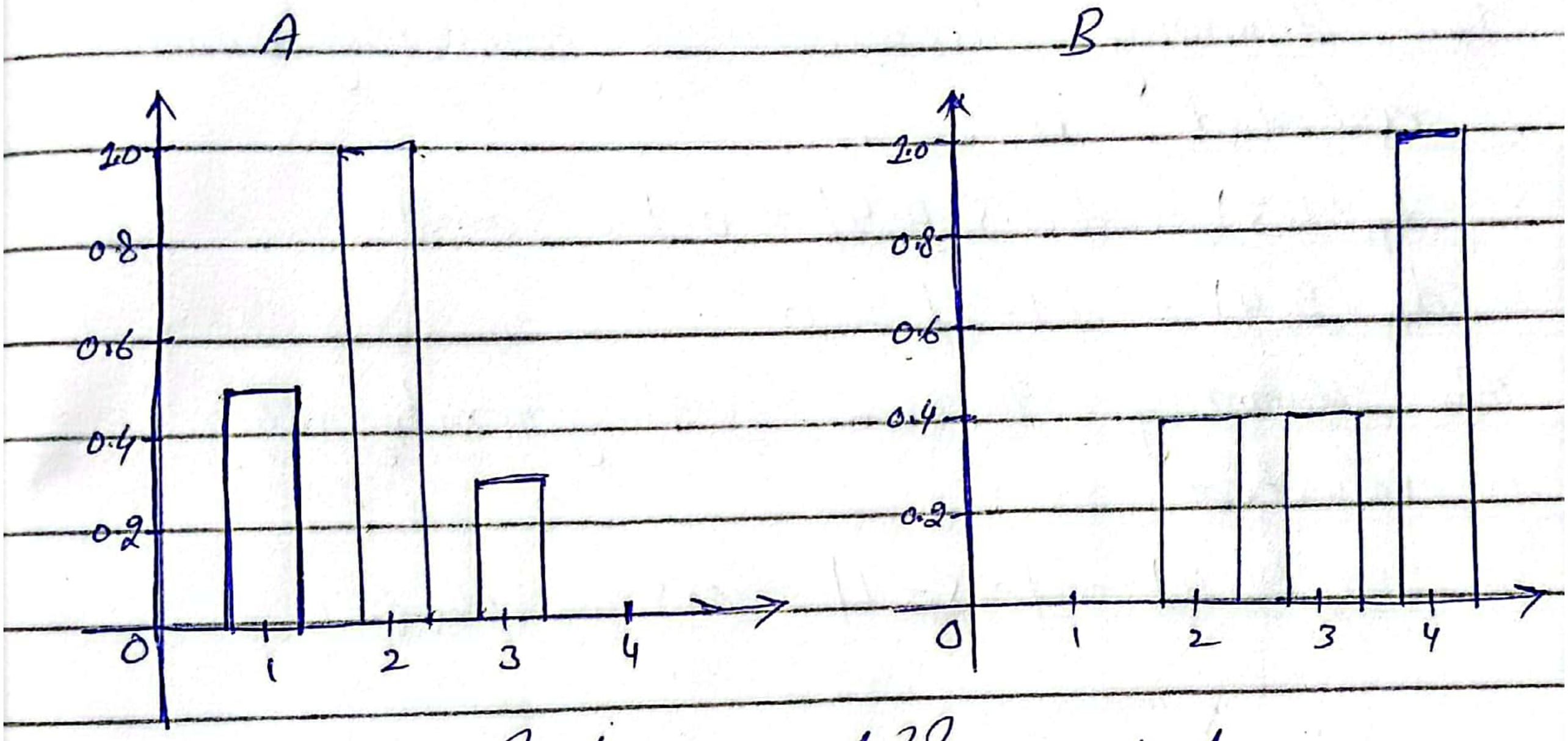
Find and plot the Hamming distance $d_h(A,B)$ between two fuzzy sets A and B on

the discourse set $X = \{1, 2, 3, 4\}$, such that

$A = \{(1, 0.5), (2, 1), (3, 0.3)\}$ and

$B = \{(2, 0.4), (3, 0.4), (4, 1)\}$.

Figure: Fuzzy sets A and B



First we find the difference between generating values, that is:

$|1-2| = 1$, $|1-3| = 2$, $|1-4| = 3$

$|2-2| = 0$, $|2-3| = 1$, $|2-4| = 2$

$|3-2| = 1$, $|3-3| = 0$ and $|3-4| = 1$.

we get the generating values (0, 1, 2, 3).

Now the Hamming distance with generating value 0 is evaluated as:

$d_h(2,2) = |1-0.4| = 0.6$ and

$d_h(3,3) = |0.3-0.4| = 0.1$.

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The Hamming distance with generating value 1 is evaluated as:

$d_h(1,2) = |0.5 - 0.4| = 0.1$

$d_h(2,3) = |1 - 0.4| = 0.6$ and

$d_h(3,4) = |0.3 - 1| = 0.7$

The Hamming distance with generating value 2 is evaluated as:

$d_h(1,3) = |0.5 - 0.4| = 0.1$ and

$d_h(2,4) = |1 - 1| = 0$

The Hamming distance with generating value 3 is evaluated as:

$d_h(1,4) = |0.5 - 1| = 0.5$, that is

$$D = 0.6 + 0.1 + 0.1 + 0.6 + 0.7 + 0.1 + 0 + 0.5$$

$$D = 2.7$$

Now the Euclidean distance with generating value 0 is evaluated as:

$$d_e(2,2) + d_e(3,3) = \sqrt{(1-0.4)^2 + (0.3-0.4)^2}$$

$$= \sqrt{(0.6)^2 + (-0.1)^2}$$

$$= 0.608$$

The Euclidean distance with generating value 1 is evaluated as:

$$d_e(1,2) + d_e(2,3) + d_e(3,4) = \sqrt{(0.5-0.4)^2 + (1-0.4)^2 + (0.3-1)^2}$$

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$$= \sqrt{(0.1)^2 + (0.6)^2 + (-0.7)^2}$$

$$= 0.9273$$

The Euclidean distance with generating value 2 is evaluated as:

$$d_e(1,3) + d_e(2,4) = \sqrt{(0.5-0.4)^2 + (1-1)^2}$$

$$= \sqrt{(0.1)^2 + (0)^2}$$

$$= 0.1$$

The Euclidean distance with generating value 3 is evaluated as:

$$d_e(1,4) = \sqrt{(0.5-1)^2} = \sqrt{(-0.5)^2}$$

$= 0.5$, that is

$$D = 0.608 + 0.9273 + 0.1 + 0.5$$

$$D = 2.13.$$

Normalized Hamming Distance.

$$D = \frac{1}{4} \cdot 2.7 = 0.675$$

Normalized Euclidean Distance

$$D = \frac{1}{4} \cdot 2.13 = 0.5325$$

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Lecture# 33:

Distance between two fuzzy sets by using max and min operations:

$$d(A,B) = \max\left[\min\left(m_A(x_i), m_B(x_i)\right)\right]$$

Consider the $X = \{1,2,3,4\}$, such that

$A = \{(1,0.5),(2,1),(3,0.3)\}$ and

$B = \{(2,0.4),(3,0.4),(4,1)\}$

Distance between fuzzy set A and B

d	x_A	x_B	$m_A(x_i)$	$m_B(x_i)$	$\min(m_A(x_i), m_B(x_i))$	$\max[\min(m_A(x_i), m_B(x_i))]$
0	2	2	1.0	0.4	0.4	0.4
	3	3	0.3	0.4	0.3	–
1	1	2	0.5	0.4	0.4	0.4
	2	3	1.0	0.4	0.4	–
	3	2	0.3	0.4	0.3	–
	3	4	0.3	1.0	0.3	–
2	1	3	0.5	0.4	0.4	1.0
	2	4	1.0	1.0	1.0	–

Therefore $d(A,B) = 0.4 + 0.4 + 1.0 = 1.8$

Now find $d(B,A) = \max[\min(m_B(x_i), m_A(x_i))]$

Consider the $X = \{1,2,3,4\}$, such that

$B = \{(2,0.4),(3,0.4),(4,1)\}$ and

$A = \{(1,0.5),(2,1),(3,0.3)\}$.

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$\mu = m\mu$

$\nu = n\mu$

Distance between fuzzy set B and A.

d	x_B	x_A	$m_B(x_i)$	$m_A(x_i)$	$\min(m_B(x_i), m_A(x_i))$	$\max[\min(---)]$
0	2	2	0.4	1.0	0.4	0.4
	3	3	0.4	0.3	0.3	–
1	2	1	0.4	0.5	0.4	0.4
	3	2	0.4	1.0	0.4	–
	2	3	0.4	0.3	0.3	–
	4	3	1.0	0.3	0.3	–
2	3	1	0.4	0.5	0.4	1.0
	4	2	1.0	1.0	1.0	–

Therefore $d(B,A) = 0.4 + 0.4 + 1.0 = 1.8$

Some Important Facts:

1. Let $X = \{x_1, x_2, \ldots, x_n\}$ be a universe set, then $D(A,B)$ the distance between two Intuitionistic fuzzy sets (IFSs) A and B satisfy the following relationship:

(i) $D_h(A,B) = 2d_h(A,B)$

(ii) $D_h^n(A,B) = 2d_h^n(A,B)$

(i) $D_h(A,B) = 2d_h(A,B)$

Let $A(\mu_A(x), \nu_A(x))$ and $B(\mu_B(x), \nu_B(x))$ be two intuitionistic fuzzy sets (IFSs), then

$$D_h(A,B) = \sum_{i=1}^{n} |\mu_A(x) - \mu_B(x)| + |\nu_A(x) - \nu_B(x)|$$

$$= \sum_{i=1}^{n} |\mu_A(x) - \mu_B(x)| + |1 - \mu_A(x) - (1 - \mu_B(x))|$$

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$$D_h(A,B) = \sum_{i=1}^{n} |\mu_A(x) - \mu_B(x)| + |-\mu_A(x) + \mu_B(x)|$$

$$D_h(A,B) = \sum_{i=1}^{n} |\mu_A(x) - \mu_B(x)| + |\mu_A(x) - \mu_B(x)|$$

$$D_h(A,B) = 2 \sum_{i=1}^{n} |\mu_A(x) - \mu_B(x)|$$

$$D_h(A,B) = 2\, d_h(A,B).$$

(ii) $D_h^n(A,B) = 2\, d_h^n(A,B)$

Let $A(\mu_A(x), \nu_A(x))$ and $B(\mu_B(x), \nu_B(x))$ be two IFSs then,

$$D_h^n(A,B) = \frac{1}{n} \sum_{i=1}^{n} |\mu_A(x) - \mu_B(x)| + |\nu_A(x) - \nu_B(x)|$$

$$= \frac{1}{n} \sum_{i=1}^{n} |\mu_A(x) - \mu_B(x)| + |1 - \mu_A(x) - (1 - \mu_B(x))|$$

$$= \frac{1}{n} \sum_{i=1}^{n} |\mu_A(x) - \mu_B(x)| + |-\mu_A(x) + \mu_B(x)|$$

$$= \frac{1}{n} \sum_{i=1}^{n} |\mu_A(x) - \mu_B(x)| + |\mu_A(x) - \mu_B(x)|$$

$$= \frac{2}{n} \sum_{i=1}^{n} |\mu_A(x) - \mu_B(x)|$$

$$= 2 \cdot \frac{1}{n} \sum_{i=1}^{n} |\mu_A(x) - \mu_B(x)|$$

$$D_h^n(A,B) = 2\, d_h^n(A,B)$$

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Lecture #34

Properties of Distance

Theorem:

Let $X = \{x_1, x_2, x_3, \ldots, x_n\}$ be a universe set, then $d(A,B)$, the distance measure between two fuzzy sets (FSs) A and B satisfy the following conditions:

(1) $0 \leq d(A,B) \leq 1$

(2) $d(A,B) = 0$ iff $A = B$

(3) $d(A,B) = d(B,A)$

(4) $d(A,C) \geq d(A,B)$ and $d(A,C) \geq d(B,C)$

for any $A, B, C \in FS(X)$.

Lecture #35

Lecture #34 and Lecture #35 are same.

Lecture #36

Intuitionistic fuzzy sets:

An intuitionistic fuzzy sets (IFS) on X is given below:

$$A = \{\langle x_i, \mu_A(x), \nu_A(x)\rangle \mid x_i \in X\}$$

where $\mu_A(x) : X \longrightarrow [0,1]$ and $\nu_A(x) : X \longrightarrow [0,1]$ satisfied the following condition,

$$0 \leq \mu_A(x) + \nu_A(x) \leq 1, \quad \forall x \in X$$

The numbers $\mu_A(x), \nu_A(x) \in [0,1]$ denote the

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degree of membership and non-membership of x to A respectively:

Some operations on Intuitionistic fuzzy sets:

Let $A(\mu_A(x), \nu_A(x))$ and $B(\mu_B(x), \nu_B(x))$ be two IFSs then

(1) $A \subseteq B$ iff $\mu_A(x) \leq \mu_B(x)$ and $\nu_A(x) \geq \nu_B(x), \forall x \in X$

(2) $A = B$ iff $\mu_A(x) = \mu_B(x)$ and $\nu_A(x) = \nu_B(x), \forall x \in X$

(3) $A \cup B = \{\langle x, \max(\mu_A(x), \mu_B(x)), \min(\nu_A(x), \nu_B(x))\rangle \mid x \in X\}$

(4) $A \cap B = \{\langle x, \min(\mu_A(x), \mu_B(x)), \max(\nu_A(x), \nu_B(x))\rangle \mid x \in X\}$

(5) $A \oplus B = \{\langle x, \mu_A(x) + \mu_B(x) - \mu_A(x) \times \mu_B(x), \nu_A(x) \times \nu_B(x)\rangle \mid x \in X\}$

(6) $A \otimes B = \{\langle x, \mu_A(x) \times \mu_B(x), \nu_A(x) + \nu_B(x) - \nu_A(x) \times \nu_B(x)\rangle \mid x \in X\}$

(7) $A - B = \{\langle x, \min(\mu_A(x), \nu_B(x)), \max(\nu_A(x), \mu_B(x))\rangle \mid x \in X\}$

(8) $A^c = \{\langle x, (\nu_A(x), \mu_A(x))\rangle \mid x \in X\}$

Example:

Let A and B be two IFSs in a nonempty set X, then,

(i) $A - B = A \cap B^c$

(ii) $A - B = B - A$ iff $A = B$

(iii) $A - B = B^c - A^c$

Sol: Let $A(\mu_A(x), \nu_A(x))$ and $B(\mu_B(x), \nu_B(x))$ be two IFSs then.

(i) $A - B = \{\langle x, \min(\mu_A(x), \nu_B(x)), \max(\nu_A(x), \mu_B(x))\rangle \mid x \in X\}$

$B^c = \{\langle x, (\nu_B(x), \mu_B(x))\rangle \mid x \in X\}$

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$A \cap B^c = \{\langle x, (\mu_A(x), \nu_A(x)) \cap (\nu_B(x), \mu_B(x)) \rangle | x \in X\}$

$A \cap B^c = \{\langle x, \min(\mu_A(x), \nu_B(x)), \max(\nu_A(x), \mu_B(x)) \rangle | x \in X\}$

$A \cap B^c = A - B$ (proved)

(ii) $A - B = \{\langle x, \min(\mu_A(x), \nu_B(x)), \max(\nu_A(x), \mu_B(x)) \rangle | x \in X\}$

If $A = B$ then $\mu_A(x) = \mu_B(x)$ and $\nu_A(x) = \nu_B(x)$

$A - B = \{\langle x, \min(\mu_B(x), \nu_A(x)), \max(\nu_B(x), \mu_A(x)) \rangle | x \in X\}$

$A - B = B - A$ (proved)

(iii) $A - B = \{\langle x, \min(\mu_A(x), \nu_B(x)), \max(\nu_A(x), \mu_B(x)) \rangle | x \in X\}$

$A^c = \{\langle x, (\nu_A(x), \mu_A(x)) \rangle | x \in X\}$

$B^c = \{\langle x, (\nu_B(x), \mu_B(x)) \rangle | x \in X\}$

$B^c - A^c = \{\langle x, \min(\nu_B(x), \mu_A(x)), \max(\mu_B(x), \nu_A(x)) \rangle | x \in X\}$

$B^c - A^c = A - B$ (proved)

Lecture #37

Distance between two IFSs:

Example:

Let us consider two IFSs A and B in $X = \{1, 2, 3, 4, 5, 6, 7\}$, such that

$$A = \frac{(0.7, 0.3)}{1} + \frac{(0.2, 0.8)}{2} + \frac{(0.6, 0.4)}{4} + \frac{(0.5, 0.5)}{5} + \frac{(1, 0)}{6}$$

$$B = \frac{(0.2, 0.8)}{1} + \frac{(0.6, 0.4)}{4} + \frac{(0.8, 0.2)}{5} + \frac{(1, 0)}{7}$$

Find the Hamming distance between A and B.

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Soln $d_h(A,B) = \sum_{i=1}^{n} |m_A(x_i) - m_B(x_i)|$

$d_h(A,B) = \sum_{i=1}^{7} |m_A(x_i) - m_B(x_i)|$

$d_h(A,B) = |0.7-0.2| + |0.3-0.8| + |0.2-0| + |0.8-1|$
$+ |0.6-0.6| + |0.4-0.4| + |0.5-0.8| + |0.5-0.2|$
$+ |1-0| + |0-1| + |0-1| + |1-0|$

$d_h(A,B) = 0.5 + 0.5 + 0.2 + 0.2 + 0 + 0 + 0.3 + 0.3$
$+ 1 + 1 + 1 + 1 = 6$

The normalized Hamming distance is

$d_h^n(A,B) = \frac{1}{n} \cdot \sum_{i=1}^{n} |m_A(x_i) - m_B(x_i)|$

$d_h^n(A,B) = \frac{1}{7} \cdot 6 = 0.86$

The Euclidean distance A and B.

$d_e(A,B) = \sqrt{\sum_{i=1}^{n} (m_A(x_i) - m_B(x_i))^2}$

$d_e(A,B) = \sqrt{\sum_{i=1}^{7} (m_A(x_i) - m_B(x_i))^2}$

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$$de(A,B) = \sqrt{(0.7-0.2)^2+(0.3-0.8)^2+(0.2-0)^2+(0.8-0.1)^2 + (0.6-0.6)^2+(0.4-0.4)^2+(0.5-0.8)^2+(0.5-0.2)^2 + (1-0)^2+(0-1)^2+(0-1)^2+(1-0)^2}$$

$$= \sqrt{(0.5)^2+(-0.5)^2+(0.2)^2+(-0.2)^2+0^2+0^2+(-0.3)^2+(0.3)^2+(1)^2+(1)^2+(-1)^2+(1)^2}$$

$$de(A,B) = \sqrt{4.76} = 2.1817$$

The normalized Euclidean distance is

$$de^n(A,B) = \frac{1}{n} \cdot \sqrt{\sum_{i=1}^{n} | m_A(x_i) - m_B(x_i) |}$$

$$de^n(A,B) = \frac{1}{7} \cdot 2.1817$$

$$de^n(A,B) = 0.3117$$

Lecture #38

Picture Fuzzy Sets:

Let $X = \{x_1, x_2, \dots, x_n\}$ be a fixed set, a picture fuzzy set P on X is defined as:

$$P = \{\langle x, \alpha_P(x), \gamma_P(x), \beta_P(x)\rangle \mid x \in X\}$$

where $\alpha_P(x), \beta_P(x), \gamma_P(x) \in [0,1]$ are called the acceptance membership, neutral and rejection membership degrees of $x \in X$ to the set P respectively and $\alpha_P(x)$, $\gamma_P(x)$ and $\beta_P(x)$ fulfil the condition:

$$0 \leq \alpha_P(x) + \gamma_P(x) + \beta_P(x) \leq 1 \quad \forall x \in X.$$

Also $\eta_P(x) = 1 - \alpha_P(x) - \gamma_P(x) - \beta_P(x)$, then $\eta_P(x)$ is said to be a degree of refusal membership

of $x \in X$ in P. For our convenience, the picture fuzzy sets (PFSs) over a fixed set X is written as PFSs(X).

Some operations on PFSs:

let A and B be two PFSs on X, then the union, interseetion and complement are described as follows:

(1) $A \subseteq B$ iff $\alpha_A(x) \leq \alpha_B(x)$, $\gamma_A(x) \leq \gamma_B(x)$ and $\beta_A(x) \geq \beta_B(x)$ such that for all $x \in X$.

(2) $A = B$ iff $A \subseteq B$ and $B \subseteq A$.

(3) $A \cup B = \{\langle x, \max(\alpha_A(x), \alpha_B(x)), \min(\gamma_A(x), \gamma_B(x)), \min(\beta_A(x), \beta_B(x))\rangle \mid x \in X\}$.

(4) $A \cap B = \{\langle x, \min(\alpha_A(x), \alpha_B(x)), \min(\gamma_A(x), \gamma_B(x)), \max(\beta_A(x), \beta_B(x))\rangle \mid x \in X\}$.

(5) $A^c = \{\langle x, \beta_A(x), \gamma_A(x), \alpha_A(x)\rangle \mid x \in X\}$.

Lecture # 39

Example:

$P_1 = \{(x_1, (0.6, 0.2, 0.1)), (x_2, (0.4, 0.1, 0.4))\}$

$P_2 = \{(x_1, (0.7, 0.3, 0)), (x_2, (0.5, 0.2, 0.3))\}$

(i) $P_1 \subseteq P_2 = ?$

$P_1 \subseteq P_2$ if $\alpha_{P_1}(x) \leq \alpha_{P_2}(x)$, $\gamma_{P_1}(x) \leq \gamma_{P_2}(x)$ and

$\beta_{P_1}(x) \geq \beta_{P_2}(x)$

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$0.6 \leq 0.7$, $0.2 \leq 0.3$ and $0.1 \geq 0$

$0.4 \leq 0.5$, $0.1 \leq 0.2$ and $0.4 \geq 0.3$

Hence $P_1 \subseteq P_2$.

(ii) $P_2 \not\subseteq P_1$

$P_1 \neq P_2$ because $P_1 \subseteq P_2$ and $P_2 \not\subseteq P_1$

(iii) $P_1 \cup P_2 = \{\langle x, \max(\alpha_{P_1}(x), \alpha_{P_2}(x)), \min(\gamma_{P_1}(x), \gamma_{P_2}(x)),$
$\min(\beta_{P_1}(x), \beta_{P_2}(x))\rangle / x \in X\}$

$P_1 \cup P_2 = \{\langle x_1, (0.7, 0.2, 0)\rangle, \langle x_2, (0.5, 0.1, 0.3)\rangle\}$

(iv) $P_1 \cap P_2 = \{\langle x, \min(\alpha_{P_1}(x), \alpha_{P_2}(x)), \min(\gamma_{P_1}(x), \gamma_{P_2}(x)),$
$\max(\beta_{P_1}(x), \beta_{P_2}(x))\rangle / x \in X\}$

$P_1 \cap P_2 = \{\langle x_1, (0.6, 0.2, 0.1)\rangle, \langle x_2, (0.4, 0.1, 0.4)\rangle\}$

(v) $P_1^c = \{\langle x, \beta_{P_1}(x), \gamma_{P_1}(x), \alpha_{P_1}(x)\rangle / x \in X\}$

$P_1^c = \{\langle x_1, (0.1, 0.2, 0.6)\rangle, \langle x_2, (0.4, 0.1, 0.4)\rangle\}$

$P_2^c = \{\langle x_1, (0, 0.3, 0.7)\rangle, \langle x_2, (0.3, 0.2, 0.5)\rangle\}$

Lecture # 40

Distance between two PFSs:

Let P_1 and P_2 be two PFSs, such that

$P_1 = \{\langle x_i, (\alpha_{P_1}(x_i), \gamma_{P_1}(x_i), \beta_{P_1}(x_i))\rangle\}$

$P_2 = \{\langle x_i, (\alpha_{P_2}(x_i), \gamma_{P_2}(x_i), \beta_{P_2}(x_i))\rangle\}$

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$\Rightarrow P_1 = (\alpha_{P_1}(x_i), \gamma_{P_1}(x_i), \beta_{P_1}(x_i))$

$P_2 = (\alpha_{P_2}(x_i), \gamma_{P_2}(x_i), \beta_{P_2}(x_i))$

Normalized Hamming distance.

$$d_h^n(A,B) = \frac{1}{n}\sum_{i=1}^{n} |m_A(x_i) - m_B(x_i)|$$

$$d_h^n(P_1,P_2) = \frac{1}{n}\sum_{i=1}^{n}\Big[|\alpha_{P_1}(x_i) - \alpha_{P_2}(x_i)| + |\gamma_{P_1}(x_i) - \gamma_{P_2}(x_i)| + |\beta_{P_1}(x_i) - \beta_{P_2}(x_i)|\Big]$$

Example: Let P_1 and P_2 be two PFSs defined on $X = \{x_1, x_2, x_3, x_4\}$, such that

$P_1 = \{\langle x_1, (0.56, 0.34, 0.1)\rangle, \langle x_2, (0.9, 0.07, 0.03)\rangle, \langle x_3, (0.4, 0.33, 0.19)\rangle, \langle x_4, (0.08, 0.79, 0.03)\rangle\}$

$P_2 = \{\langle x_1, (0.7, 0.1, 0.09)\rangle, \langle x_2, (0.1, 0.66, 0.2)\rangle, \langle x_3, (0.06, 0.81, 0.12)\rangle, \langle x_4, (0.72, 0.14, 0.08)\rangle\}$

$d_h^n(P_1,P_2) = ?, \quad n = 4$

$d_h^n(P_1,P_2) = \frac{1}{4}[|0.56-0.7| + |0.34-0.1| + |0.1-0.09| + |0.9-0.1| + |0.07-0.66| + |0.03-0.2| + |0.4-0.06| + |0.33-0.81| + |0.19-0.12| + |0.08-0.72| + |0.79-0.14| + |0.03-0.08|]$

$= \frac{1}{4}[0.14 + 0.24 + 0.01 + 0.8 + 0.59 + 0.17 + 0.34 + 0.48 + 0.07 + 0.63 + 0.65 + 0.06]$

$d_h^n(P_1,P_2) = \frac{1}{4}[4.18] = 1.045$ Ans

Let $P_1 = (\alpha_{P_1}(x_i), \gamma_{P_1}(x_i), \beta_{P_1}(x_i))$ and

$P_2 = (\alpha_{P_2}(x_i), \gamma_{P_2}(x_i), \beta_{P_2}(x_i))$ be two PFSs

defined on $X = \{x_1, x_2, \ldots, x_n\}$

$$D_p(P_1, P_2) = \frac{1}{3n} \sum_{i=1}^{n} \Big[|\alpha_{P_1}(x_i) - \alpha_{P_2}(x_i)| + |\gamma_{P_1}(x_i) - \gamma_{P_2}(x_i)| + |\beta_{P_1}(x_i) - \beta_{P_2}(x_i)| + \max\big[|\alpha_{P_1}(x_i) - \alpha_{P_2}(x_i)|, |\gamma_{P_1}(x_i) - \gamma_{P_2}(x_i)|, |\beta_{P_1}(x_i) - \beta_{P_2}(x_i)|\big] \Big]$$

Example: let P_1 and P_2 be two PFSs defined on $X = \{x_1, x_2, x_3\}$, such that

$P_1 = \{\langle x_1, (0.8, 0.1, 0)\rangle, \langle x_2, (0.4, 0.2, 0.3)\rangle, \langle x_3, (0.5, 0.3, 0)\rangle\}$

$P_2 = \{\langle x_1, (0.3, 0.3, 0.2)\rangle, \langle x_2, (0.7, 0.1, 0.1)\rangle, \langle x_3, (0.4, 0.3, 0.2)\rangle\}$

$D_p(P_1, P_2) = ?, \quad n = 3$

$$D_p(P_1, P_2) = \frac{1}{3(3)} \Big[|0.8-0.3| + |0.1-0.3| + |0-0.2| + \max\big[|0.8-0.3|, |0.1-0.3|, |0-0.2|\big] + |0.4-0.7| + |0.2-0.1| + |0.3-0.1| + \max\big[|0.4-0.7|, |0.2-0.1|, |0.3-0.1|\big] + |0.5-0.4| + |0.3-0.3| + |0-0.2| + \max\big[|0.5-0.4|, |0.3-0.3|, |0-0.2|\big] \Big]$$

$$D_p(P_1, P_2) = \frac{1}{9} \Big[0.5 + 0.2 + 0.2 + 0.5 + 0.3 + 0.1 + 0.2 + 0.3 + 0.1 + 0 + 0.2 + 0.2 \Big]$$

$$= \frac{1}{9}[2.8] = 0.311 \quad \text{Ans}$$

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Lecture #41

Hasitant Fuzzy Sets (HFSs):

Let X be a reference set, then we define a hesitant fuzzy set on X in terms of a function h that when applied to X returns a subset of [0,1]. A HFS H on X is written as:

$$H = \{\langle x, h(x)\rangle \mid \forall x \in X\}$$

where $h(x)$ is a set of some different values in [0,1], representing the possible membership degrees of the element $x \in X$ to H.

Lecture #42

Example:

$$X = \{x_1, x_2\}$$

$$H_1 = \{\langle x_1, \{0.1, 0.2\}\rangle, \langle x_2, \{0.5, 0.6, 0.8\}\rangle\}$$

$$H_2 = \{\langle x_1, \{0.8\}\rangle, \langle x_2, \{0.6, 0.7\}\rangle\}$$

Some Operations on HFSs:

For any h, h_1 and h_2 in $H(X)$, some operations on them can be described as follows:

(1) Lower bound: $h^-(x) = \min h(x)$;

(2) Upper bound: $h^+(x) = \max h(x)$;

(3) Complement: $h^c(x) = \{1 - \gamma \mid \gamma \in h(x)\}$

(4) Union: $(h_1 \cup h_2)(x) = \{\gamma \in h_1 \cup h_2 \mid \gamma \geq \max(h_1^-(x), h_2^-(x))\}$.

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(5) Intersection: $(h_1 \cap h_2)(x) = \{\gamma \in h_1^{(x)} \cap h_2^{(x)} \mid \gamma \leq \min(h_1^+(x), h_2^+(x))\}$;

(6) $h^\lambda(x) = \{\gamma^\lambda \mid \gamma \in h(x)\}$;

(7) $\lambda h(x) = \{1 - (1-\gamma)^\lambda \mid \gamma \in h(x)\}$;

(8) $(h_1 \oplus h_2)(x) = \{\gamma_1 + \gamma_2 - \gamma_1 \cdot \gamma_2 \mid \gamma_1 \in h_1(x), \gamma_2 \in h_2(x)\}$;

(9) $(h_1 \otimes h_2)(x) = \{\gamma_1 \cdot \gamma_2 \mid \gamma_1 \in h_1(x), \gamma_2 \in h_2(x)\}$.

Lecture # 43

Example: Let $X = \{x_1, x_2, x_3\}$ be a universal set. h_1 and h_2 be two HFSs on X such that

$h_1 = \{\langle x_1, \{0.3, 0.4\}\rangle, \langle x_2, \{0.6, 0.8\}\rangle, \langle x_3, \{0.3, 0.4, 0.5, 0.7\}\rangle\}$

$h_2 = \{\langle x_1, \{0.5, 0.6\}\rangle, \langle x_2, \{0.4, 0.5\}\rangle, \langle x_3, \{0.2, 0.3, 0.4, 0.6\}\rangle\}$

Then find out the following operations:-

(i) $h_1^-(x_3) = ?$

$h_1^-(x_3) = \min\{0.3, 0.4, 0.5, 0.7\} = 0.3$

(ii) $h_2^+(x_2) = ?$

$h_2^+(x_2) = \max\{0.4, 0.5\} = 0.5$

(iii) $h_1^c(x_2) = ?$ $\qquad h_1^c(x) = \{1 - \gamma \mid \gamma \in h\}$

$h_1^c(x_2) = \{1 - 0.6, 1 - 0.8\} = \{0.4, 0.2\}$

(iv) $h_1^\lambda(x_2) = ?$ if $\lambda = 2$ $\qquad h^\lambda(x) = \{\gamma^\lambda\}$

$h_1^\lambda(x_2) = \{0.6^2, 0.8^2\} = \{0.36, 0.64\}$

(v) $\lambda h_1(x_1) = ?$ if $\lambda = 2$ $\qquad \lambda h(x) = \{1 - (1-\gamma)^\lambda\}$

$\lambda h_1(x_1) = \{1 - (1-0.3)^2, 1 - (1-0.4)^2\}$

$= \{1 - 0.7^2, 1 - 0.6^2\} = \{1 - 0.49, 1 - 0.36\}$

$\lambda h_1(x_1) = \{0.51, 0.64\}$

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(vi) $(h_1 \oplus h_2)(x_2) = ?$ $\quad (h_1 \oplus h_2)(x) = \{\gamma_1 + \gamma_2 - \gamma_1\gamma_2\}$

$(h_1 \oplus h_2)(x_2) = \{0.6, 0.8\} \oplus \{0.4, 0.5\}$

$= \{0.6+0.4-(0.6)(0.4),\ 0.6+0.5-(0.6)(0.5),\ 0.8+0.4-(0.8)(0.4),$

$0.8+0.5-(0.8)(0.5)\}$

$= \{1.0-0.24,\ 1.1-0.30,\ 1.2-0.32,\ 1.3-0.40\}$

$= \{0.76,\ 0.8,\ 0.88,\ 0.9\}$

(vii) $(h_1 \otimes h_2)(x_1) = ?$ $\quad (h_1 \otimes h_2)(x) = \{\gamma_1 \cdot \gamma_2\}$

$(h_1 \otimes h_2)(x_1) = \{0.3, 0.4\} \otimes \{0.5, 0.6\}$

$= \{(0.3)(0.5),\ (0.3)(0.6),\ (0.4)(0.5),\ (0.4)(0.6)\}$

$= \{0.15,\ 0.18,\ 0.20,\ 0.24\}$

(viii) $(h_1 \cup h_2)(x_3) = ?$

$(h_1 \cup h_2)(x_3) \geq \max\{\min\{0.3, 0.4, 0.5, 0.7\}, \min\{0.2, 0.3, 0.4, 0.6\}\}$

$\geq \max\{0.3, 0.2\} = 0.3$

$= \{0.3, 0.4, 0.5, 0.6, 0.7\}$

(ix) $(h_1 \cap h_2)(x_3) = ?$

$(h_1 \cap h_2)(x_3) \leq \min\{\max\{0.3, 0.4, 0.5, 0.7\}, \max\{0.2, 0.3, 0.4, 0.6\}\}$

$\leq \min\{0.7, 0.6\} = 0.6$

$= \{0.2, 0.3, 0.4, 0.5, 0.6\}$

Lecture #44

Hesitant Fuzzy Element (HFE):

$h_1 = \{\langle x_1, \{0.2, 0.3\}\rangle\} \Rightarrow h_1 = (0.2, 0.3) \rightarrow$ (HFE)

Theorem:

Suppose that h_1 and h_2 are two HFEs, then

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$$L_{h_1 \oplus h_2} = L_{h_1} * L_{h_2} \quad \text{and}$$

$$L_{h_1 \otimes h_2} = L_{h_1} \times L_{h_2}$$

Example: Let $h_1 = (0.1, 0.2, 0.7)$ and $h_2 (0.2, 0.4)$ be two HFEs then

$$h_1 \oplus h_2 = \{0.1, 0.2, 0.7\} \oplus \{0.2, 0.4\}$$

$$= \{0.1 + 0.2 - (0.1)(0.2),\ 0.1 + 0.4 - (0.1)(0.4),\ 0.2 + 0.2 - (0.2)(0.2),$$

$$0.2 + 0.4 - (0.2)(0.4),\ 0.7 + 0.2 - (0.7)(0.2),\ 0.7 + 0.4 - (0.7)(0.4)\}$$

$$= \{0.28, 0.46, 0.36, 0.52, 0.76, 0.82\}$$

$$h_1 \otimes h_2 = \{0.1, 0.2, 0.7\} \otimes \{0.2, 0.4\}$$

$$= \{(0.1)(0.2), (0.1)(0.4), (0.2)(0.2), (0.2)(0.4), (0.7)(0.2), (0.7)(0.4)\}$$

$$= \{0.02, 0.04, 0.04, 0.08, 0.14, 0.28\}$$

$$\therefore L_{h_1 \oplus h_2} = 6 = 3 \times 2 = L_{h_1} \times L_{h_2}$$

$$L_{h_1 \otimes h_2} = 6 = 3 \times 2 = L_{h_1} \times L_{h_2} \quad \text{(proved)}$$

Distance between HFSs:

$$De(h_1, h_2) = \sqrt{\sum_{i=1}^{n} (h_{1i} - h_{2i})^2}$$

$$D_h(h_1, h_2) = \sum_{i=1}^{n} |h_{1i} - h_{2i}|$$

$$De^n(h_1, h_2) = \frac{1}{n} \sqrt{\sum_{i=1}^{n} (h_{1i} - h_{2i})^2}$$

$$D_h^n(h_1, h_2) = \frac{1}{n} \sum_{i=1}^{n} |h_{1i} - h_{2i}|$$

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Example1: Let $h_1 = (0.7, 0.8, 0.9)$ and $h_2 = (0.5, 0.6, 0.8)$ be two HFEs, then evaluate (i) $D_h(h_1, h_2)$ (ii) $D_e(h_1, h_2)$

Sol: (i) $D_h(h_1, h_2) = |0.7 - 0.5| + |0.8 - 0.6| + |0.9 - 0.8|$

$= 0.2 + 0.2 + 0.1 = 0.5$

(ii) $D_e(h_1, h_2) = \sqrt{(0.7-0.5)^2 + (0.8-0.6)^2 + (0.9-0.8)^2}$

$= \sqrt{(0.2)^2 + (0.2)^2 + (0.1)^2} = \sqrt{0.04 + 0.04 + 0.01} = \sqrt{0.09}$

$= 0.3$

Example 2: Let $h_1 = (0.3, 0.4)$ and $h_2 = (0.6, 0.7, 0.8)$ be two HFEs, then evaluate the normalized Hamming distance and normalized Euclidean distance between h_1 and h_2.

Note: (Add minimum element for normalized HFEs)

Sol: $h_1 = (0.3, 0.3, 0.4)$, $h_2 = (0.6, 0.7, 0.8)$

$D_h^n(h_1, h_2) = \frac{1}{3}[|0.3 - 0.6| + |0.3 - 0.7| + |0.4 - 0.8|]$

$= \frac{1}{3}[0.3 + 0.4 + 0.4] = \frac{1}{3}(1.1)$

$D_h^n(h_1, h_2) = 0.367$

$D_e^n(h_1, h_2) = \frac{1}{3}\sqrt{(0.3-0.6)^2 + (0.3-0.7)^2 + (0.4-0.8)^2}$

$= \frac{1}{3}\sqrt{(0.3)^2 + (0.4)^2 + (0.4)^2}$

$= \frac{1}{3}\sqrt{0.09 + 0.16 + 0.16}$

$= \frac{1}{3}\sqrt{0.41} = \frac{1}{3}(0.64)$

$D_e^n(h_1, h_2) = 0.213$

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Lecture # 15

Similarity:

Two objects are said to be similar if both have same shape.

e.g. Two bags have same shape and ~~not~~ have not same size.

Similarity Measure (SM):

Similarity measure is a process through which we can evaluate resamblence between two objects.

There are many kinds of similarity measures like

(i) similarity measures based on distance.

(ii) Cosine similarity measures.

(iii) Exponential similarity measures.

(iv) Jaccard and Dice similarity measures.

(v) William and Steel similarity measures. etc.

Similarity Measures based on distance:

Let S be a Similarity Measures between two fuzzy sets is calculated as:-

Suppose that A and B are two fuzzy sets then the similarity measure S(A,B) based on distance. $d_h(A,B)$ is written as:

$$S(A,B) = 1 - d_h(A,B)$$

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or $S(A,B) = 1 - de(A,B)$

or $S(A,B) = 1 - de^n(A,B)$

or $S(A,B) = 1 - d_n^n(A,B)$

Similarity Measures satisfied the condition

$0 \leq S(A,B) \leq 1$

Theorem: The similarity measures $S(A,B)$ between two fuzzy sets satisfied the following axioms:-

(i) $0 \leq S(A,B) \leq 1$

(ii) $S(A,B) = 1$ if and only if $A = B$

(iii) $S(A,B) = S(B,A)$ i.e commutative.

End

Always remember in prayers.

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